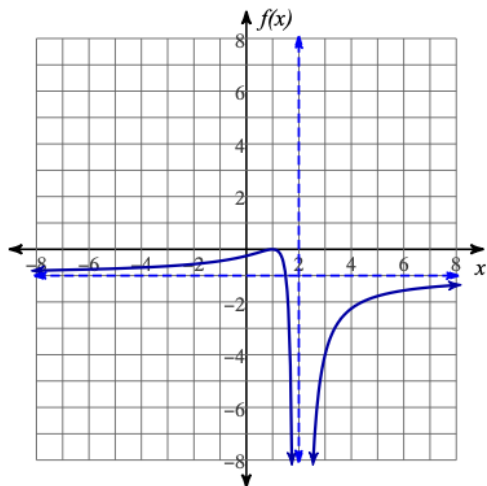


State the open intervals where the function is increasing and decreasing and also state the **NUMBER** of critical values that exist in this function. The equation is only there for information, solve this just by looking at the graph. (4 points)

1. $f(x) = -\left(\frac{x-1}{x-2}\right)^2$



For the problem, find the absolute minimum and maximum points over the given interval. (5 points)

2. $y = -\frac{x^2}{2} - 3x - \frac{11}{2}$

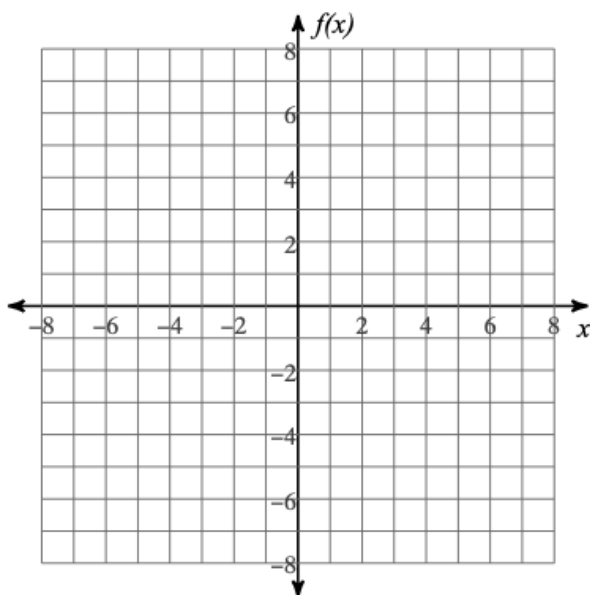
For each problem, find all local max and min points. Use the first derivative test for #3 and the second derivative test for #4. (8 points)

3. $y = -\frac{3}{x^2-9}$

4. $y = -x^3 + 4x^2 - 2$

For this problem, find the local max and min points, the open intervals of increase and decrease, find points of inflection, and find the open intervals where the function is concave up and concave down. Also identify the y-intercept. Using this information, sketch a graph of the function on the grid provided. (10 points)

5. $f(x) = x^3 - x^2 - 5x - 1$



Determine if the Mean Value Theorem can be applied. If it can, find all values of c that satisfy the theorem. If it cannot, explain why not. (6 points)

6. $f(x) = x^2 + 6x + 10$; $[-5, -2]$

7. $f(x) = \frac{x^2}{2x-4}$; $[0, 3]$

8. A certain function exists such that $f(-4) = 3$, $f'(-4) = 2$, and $f''(-4) = 0$. From this information, in a sentence or two, determine the nature of the point at $f(-4)$. Draw a small sketch of what the curve looks like "near" $f(-4)$. (3 points)

9. For any cubic (3rd degree) polynomial, what are the least and most number of inflection points that the curve could have? (2 points)

10. If $f(x)$ is a continuous function and $(c, f(c))$ is a critical point of $f(x)$, when is there no local maximum or minimum at $x = c$? EXPLAIN in a sentence how this can be possible. You may also include a sketch. (3 points)